Complex Number: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Basics, powers of i, Argand plane

Formulas

$$z = x + iy, \quad \operatorname{Re} z = x,\ \operatorname{Im} z = y, \quad i^{4k} = 1,\ i^{4k+1} = i,\ i^{4k+2} = -1,\ i^{4k+3} = -i$$

KEY POINTS

- Purely real: $\operatorname{Im} z = 0$ (real axis); purely imaginary: $\operatorname{Re} z = 0$ (imaginary axis); 0 is both
- $z = x + iy$ is the point (x, y); sum of 4 consecutive powers of i is 0
- $\sqrt{a}\sqrt{b} = \sqrt{ab}$ fails when both $a, b < 0$: $\sqrt{-2}\sqrt{-3} = -\sqrt{6}$

2. Algebra, equality, square root

Formulas

$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{c^2+d^2}, \qquad \sqrt{a+ib} = \pm\left(\sqrt{\frac{|z|+a}{2}} + i\,\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right)$$

KEY POINTS

- $z_1 = z_2 \iff$ real parts equal and imaginary parts equal
- Inequality is defined only for real numbers ($|z|$, $\operatorname{Re} z$, $\operatorname{Im} z$ are real)
- Quadratic with complex coefficients: use the formula with a complex $\sqrt{D}$; roots need not be conjugate

STEPS

1. Equation with $\bar{z}, |z|$: take modulus first (find $|z|$), or put $z = x + iy$ and equate parts
2. Count $z = 0$ separately when you divide by z

3. Conjugate

Formulas

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{(z_1/z_2)} = \bar{z}_1/\bar{z}_2, \quad z + \bar{z} = 2\operatorname{Re} z, \quad z - \bar{z} = 2i \operatorname{Im} z$$

KEY POINTS

- $\bar{z}$ = mirror image of z in the real axis
- Purely real $\iff z = \bar{z}$; purely imaginary $\iff z + \bar{z} = 0$
- Real-coefficient polynomial: non-real roots come in conjugate pairs

4. Modulus and triangle inequality

Formulas

$$|z| = \sqrt{x^2+y^2}, \quad z\bar{z} = |z|^2, \quad |z_1 z_2| = |z_1||z_2|, \quad \big||z_1| - |z_2|\big| \le |z_1 \pm z_2| \le |z_1| + |z_2|$$

$$|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\operatorname{Re}(z_1 \bar{z}_2), \qquad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

KEY POINTS

- $|z|$ = distance from O; $|z_1 - z_2|$ = distance between the points
- $|z| = 1 \iff \bar{z} = 1/z$; $|z| = r \iff \bar{z} = r^2/z$
- Equality $|z_1 + z_2| = |z_1| + |z_2|$: same direction ($\arg z_1 = \arg z_2$); $\big||z_1| - |z_2|\big|$: opposite directions
- Max / min of $|z - z_0|$ on a circle: $|z_0 - c| \pm r$

Watch out

$|z_1 + z_2| \neq |z_1| + |z_2|$ in general; $z^2 \neq |z|^2$ unless z is real

5. Argument

Formulas

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2, \quad \arg\frac{z_1}{z_2} = \arg z_1 - \arg z_2, \quad \arg z^n = n \arg z, \quad \arg \bar{z} = -\arg z \quad (+2k\pi)$$

KEY POINTS

- Principal argument in $(-\pi, \pi]$; quadrant 1st: α, 2nd: $\pi - \alpha$, 3rd: $-\pi + \alpha$, 4th: $-\alpha$ with $\alpha = \tan^{-1}|b/a|$
- Axes: 0, π, $\pi/2$, $-\pi/2$; $\arg 0$ is not defined
- $\arg z = \theta$ is a ray from O (origin excluded)

Watch out

After adding arguments bring the answer back into $(-\pi, \pi]$

6. Polar and exponential form

Formulas

$$z = r(\cos\theta + i\sin\theta) = r\,\mathrm{CiS}\,\theta = re^{i\theta}, \qquad e^{i\theta} = \cos\theta + i\sin\theta$$

$$1 + \cos\theta + i\sin\theta = 2\cos\tfrac{\theta}{2}e^{i\theta/2}, \quad 1 - e^{i\theta} = -2i\sin\tfrac{\theta}{2}e^{i\theta/2}, \quad 1 + i\tan\theta = \sec\theta\, e^{i\theta}$$

KEY POINTS

- Look-alike forms: $\cos\theta - i\sin\theta = e^{-i\theta}$, $\sin\theta + i\cos\theta = e^{i(\pi/2-\theta)}$
- Exponential form for products, powers, roots, rotation; polar form when cos / sin are compared
- $|z| = 1$: put $z = e^{i\theta}$; then $\dfrac{1+z}{1+\bar{z}} = z$ and $\dfrac{1-z}{1+z} = -i\tan\frac{\theta}{2}$

Watch out

Modulus must be positive: if $2\cos\frac{\theta}{2} < 0$, write $-2\cos\frac{\theta}{2}$ and add π to the angle

7. Geometry: distance, section, triangle

Formulas

$$z = \frac{mz_2 \pm nz_1}{m \pm n}, \quad G = \frac{z_1 + z_2 + z_3}{3}, \quad I = \frac{az_1 + bz_2 + cz_3}{a+b+c}$$

$$H = \frac{\sum z_1 \tan A}{\sum \tan A}, \quad O = \frac{\sum z_1 \sin 2A}{\sum \sin 2A}$$

KEY POINTS

- $z_2 - z_1 =$ vector $\vec{AB}$
- $0, z_1, z_2, z_1 + z_2$: parallelogram, diagonals $z_1 \pm z_2$; $z_1 + z_3 = z_2 + z_4$ for a parallelogram $z_1 z_2 z_3 z_4$
- Circumcentre at $O \Rightarrow$ orthocentre $= z_1 + z_2 + z_3$; $\sum |z - z_k|^2$ least at the centroid

8. Loci

Formulas

$$|z - z_0| = r, \qquad |z - z_1| = |z - z_2|, \qquad \left|\frac{z - z_1}{z - z_2}\right| = k \neq 1$$

KEY POINTS

- These are: circle; perpendicular bisector of $z_1 z_2$; circle (Apollonius)
- $|z - z_1| + |z - z_2| = k$: ellipse ($k > |z_1 - z_2|$), segment ($=$), none ($<$)
- $\big||z - z_1| - |z - z_2|\big| = k$: hyperbola ($k < |z_1 - z_2|$), two rays ($=$), none ($>$)
- $|z - a| = \mathrm{Re}(z) + c$ type: parabola; $\mathrm{Re}(z^2) = c$: rectangular hyperbola

STEPS

1. Recognise a standard figure first (distance / angle reading)
2. Otherwise put $z = x + iy$ and simplify to a real equation
3. For max / min use the geometry (line through the centre, tangent from the origin)

9. Angles and rotation

Formulas

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\theta}, \qquad \theta = \arg \frac{z_3 - z_1}{z_2 - z_1}$$

KEY POINTS

- Multiply by $e^{i\theta}$: rotate θ anticlockwise; by i: 90°; by ω: 120°; clockwise angles are negative
- Parallel / collinear: ratio of the two vectors real; perpendicular: ratio purely imaginary
- $\arg \dfrac{z - z_1}{z - z_2} = \theta$: arc of a circle through z_1, z_2; $\pm\dfrac{\pi}{2}$: circle on diameter $z_1 z_2$
- Equilateral: $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$; regular n-gon: $z_k = z_0 + (z_1 - z_0) e^{i2\pi(k-1)/n}$

Watch out

Both vectors must start at the vertex of the angle; check the order (anticlockwise) of the vertices

10. De Moivre's theorem and nth roots

Formulas

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta \ (n \in \mathbb{Z}), \qquad z^{1/n} = r^{1/n} e^{i(\theta + 2k\pi)/n}, \ k = 0, \ldots, n-1$$

STEPS

1. Write $z = re^{i\theta}$
2. Replace θ by $\theta + 2k\pi$
3. Apply the power $1/n$ and put $k = 0, 1, \ldots, n-1$

KEY POINTS

- nth roots lie on $|z| = r^{1/n}$ at a regular n-gon
- Real / purely imaginary root: put $z = x$ (or iy); real and imaginary parts both 0 (common root)
- $\alpha^n + \beta^n$ for complex roots: write $\alpha, \beta = re^{\pm i\theta}$, get $2r^n \cos n\theta$

11. Cube roots of unity

Formulas

$$\omega = \frac{-1 + i\sqrt{3}}{2} = e^{i2\pi/3}, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \bar{\omega} = \omega^2, \quad -\omega = e^{-i\pi/3}, \ -\omega^2 = e^{i\pi/3}$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega), \quad |a + b\omega + c\omega^2|^2 = a^2 + b^2 + c^2 - ab - bc - ca$$

KEY POINTS

- ω^n: reduce n mod 3; $1 + \omega^r + \omega^{2r} = 3$ if $3 \mid r$, else 0
- $x^2 + x + 1 = 0$: ω, ω^2; $x^2 - x + 1 = 0$: $-\omega, -\omega^2$
- $1, \omega, \omega^2$: equilateral triangle in $|z| = 1$, side $\sqrt{3}$
- $1 + \omega = -\omega^2$, $1 + \omega^2 = -\omega$; $1 \pm i\sqrt{3} = -2\omega^2, \ -2\omega$

12. nth roots of unity

Formulas

$$\alpha_k = e^{i2k\pi/n}, \quad \sum \alpha_k = 0, \quad \prod \alpha_k = (-1)^{n-1}, \quad x^n - 1 = \prod_{k=0}^{n-1} (x - \alpha_k)$$

$$\sum_k \alpha_k^r = n \ (n \mid r), \ 0 \ (n \nmid r), \qquad \prod_{k=1}^{n-1} (x - \alpha_k) = 1 + x + \cdots + x^{n-1}$$

KEY POINTS

- GP with ratio $e^{i2\pi/n}$; all on $|z| = 1$; regular n-gon of side $2\sin\frac{\pi}{n}$, area $\frac{n}{2}\sin\frac{2\pi}{n}$
- $\overline{\alpha^k} = \alpha^{n-k} = 1/\alpha^k$; real factors $x^2 - 2x\cos\frac{2k\pi}{n} + 1$
- Product of distances from a point z to the vertices $= |z^n - 1|$ (unit circle polygon)

13. Series and binomial coefficients

Formulas

$$\sum_{k=0}^{n-1}\cos(\alpha+k\beta)=\frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\cos\left(\alpha+\frac{(n-1)\beta}{2}\right)\ \text{(same with sin)}$$

$$C_0-C_2+C_4-\cdots=2^{n/2}\cos\tfrac{n\pi}{4},\quad C_0+C_3+C_6+\cdots=\tfrac{1}{3}\left(2^n+2\cos\tfrac{n\pi}{3}\right)$$

STEPS

1. Trig series: write $C+iS$ as a GP of $e^{i\theta}$, sum, take Re / Im
2. Binomial series: put $x=i$ (step 2 with signs or step 4), $x=\omega$ (step 3), $x=\alpha_p$ (step p) and average

14. Line and circle in complex form

Formulas

$$z=(1-t)z_1+tz_2;\qquad \bar{a}z+a\bar{z}+b=0\ (b\in\mathbb{R});\qquad z\bar{z}+\bar{a}z+a\bar{z}+b=0,\ C=-a,\ r=\sqrt{|a|^2-b}$$

KEY POINTS

- Image of z_1 in $\bar{a}z+a\bar{z}+b=0$: $\bar{a}z_2+a\bar{z}_1+b=0$
- Collinear: $\begin{vmatrix} z_1 & \bar{z}_1 & 1\\ z_2 & \bar{z}_2 & 1\\ z_3 & \bar{z}_3 & 1\end{vmatrix}=0$; diametric circle $|z-z_1|^2+|z-z_2|^2=|z_1-z_2|^2$
- Concyclic: $\dfrac{(z_4-z_1)(z_2-z_3)}{(z_2-z_1)(z_4-z_3)}$ real; half-plane $\bar{a}z+a\bar{z}+b\geq 0$ is a region bounded by the line

15. Common mistakes checklist

STEPS

1. Principal argument must lie in $(-\pi,\pi]$ (not $[0,2\pi)$)
2. Use the general argument $\theta+2k\pi$ before taking nth roots
3. Do not write $z_1>z_2$ for non-real numbers
4. $\arg z=\theta$ and $\arg\frac{z-z_1}{z-z_2}=\theta$ exclude the end points
5. Purely imaginary needs $z\neq 0$; check excluded points ($z\neq z_2$ in a quotient)
6. Conjugate roots only for real coefficients
7. Check the sign of the modulus in look-alike polar forms